

SCALABILITY & FUTURE PLAN

Date	30 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	Uses historical dataset only	Predictions cannot adapt to changing real-time weather conditions.	High
2	No location-based prediction	Users must manually enter weather parameters instead of selecting a location.	Medium
3	Free-tier deployment (Render)	Initial loading may be slow due to server cold start.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of concurrent users.	Deploy on a scalable cloud platform with higher computing resources and load balancing.
2	Data Storage	Uses local dataset and Joblib model files.	Integrate a cloud database and continuously update the dataset with new records.
3	Performance	Prediction is performed on a single Flask server.	Optimize the application using caching, faster inference techniques, and production-ready web servers.
4	Security	Basic input validation is implemented.	Add user authentication, HTTPS, secure API handling, and role-based access control.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Integrate live weather APIs for real-time data collection.	Version 2.0	Improve prediction accuracy using live weather information.
Phase 2	Implement location-based flood prediction using GPS and maps.	Version 2.1	Provide region-specific flood predictions for users.
Phase 3	Develop a mobile application with push notifications.	Version 3.0	Increase accessibility and provide instant flood alerts.
Phase 4	Integrate GIS visualization and interactive flood risk maps.	Version 3.5	Help users and authorities visualize flood-prone areas effectively.